Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/29/2005
PI Name:	Savary, Cherylyn A Ph.D.		
Project Title:	Study of Molecular Lesions in Dendritic Cells Cultivated in Modeled Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	csavary@mdanderson.org	**Fax:**	FY 713-563-3424
PI Organization Type:	UNIVERSITY	**Phone:**	713-792-3134
Organization Name:	The University of Texas M.D. Anderson Cancer Center		
PI Address 1:	7455 Fannin, Unit 904		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77054	**Congressional District:**	25
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Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006) NOTE: Project taken over by G. Lizee in January 2006; search that PI for information after 12/2005		
Key Personnel Changes/Previous PI:	Greg Lizee has replaced Dr. Savary as PI, effective January 2006 (per info from HQ 9/06). For information after 12/31/2005, search on Lizee and new project title, "Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity."		
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Space flight has a significant negative impact on immune function. This is a concern because microorganisms that cause life-threatening opportunistic infections on earth thrive aboard spacecraft. We have found that human dendritic cell (DC) maturation and functions were attenuated in a microgravity analog culture, i.e., the rotary cell culture system (RCCS). DCs are considered to be the most efficient of antigen-presenting cells. Because they play a central role in development of innate and acquired immune responses, a defect in DC would be expected to result in a broad impairment of immune function, not unlike that observed in astronauts during or after space flight. Yet there is nothing known about the effects of space flight on DC.
In our previous studies we observed that, when compared to DC generated in standard static culture conditions, DC grown in RCCS displayed a decreased ability to phagocytose A. fumigatus conidia, reduced levels of surface MHC class II and CD80 costimulatory molecules, and impaired production of IL-12 after exposure to the fungus. These observations suggest that DC generated in modeled microgravity lack the processing and presentation properties needed to mount an effective antifungal immune response. If a similar situation occurs in true microgravity, it could be an underlying factor contributing to the immune blunting associated with space flight, and would emphasize the need to develop effective countermeasures to protect the health of astronauts. Before countermeasures can be developed, though, we must identify targets where countermeasures could be effective. Therefore, the objectives of this study are to identify the cellular and molecular lesions in DC-genesis and function that occur in an in vitro microgravity analog.

Task Description:

We plan to: (a) identify molecular lesions in human DC maturation and function that occur in the modeled microgravity environment of the RCCS; (b) examine DC maturation and function in the murine RCCS model; and (c) evaluate the effects of dietary nucleotides on murine and human DC generation and function in the RCCS.

There is overwhelming data from space flight experiments to suggest that a defect in antigen presenting cells might occur in microgravity. Until routine testing of space travelers is possible, we must make use of ground-based models proven to mimic (to the degree that it is possible on earth) some of the effects on immune performance known to be associated with space flight. The RCCS is such a model. In fact, in our experience with this model, we have already detected several defects in DC maturation and function that could explain some of the immune blunting reported for astronauts/cosmonauts. A novel aspect of our studies has been the inclusion of Aspergillus and Candida antigens as targets for immune testing. We are particularly interested in gaining more information on host defense against Aspergillus because of the urgent need to develop new treatments for this opportunistic mould that causes life-threatening infections in immunocompromised patients. These fungi colonize spacecraft and could also impact the health of astronauts. Thus, our studies address a significant problem in space medicine. In our quest to identify lesions associated with impaired DC maturation and function in modeled microgravity, we will contribute new knowledge to the area of DC immunobiology in general. Moreover, we should be able to identify targets for development of countermeasures to buffer against these negative effects to the benefit of both astronauts and earth-bound patients.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

We have begun to explore the mechanisms responsible for low DC yield in RCCS compared to standard/static cultures. Human CD14+ monocytes were seeded into RCCS rotating vertically (V-RCCS; microgravity analog), RCCS rotating horizontally (H-RCCS; to control for sheer stress) and tissue culture flasks (static culture controls) in medium containing GM-CSF+IL-4 – conditions known to support generation of DCs. Using Annexin-V staining, we have detected a higher percentage of apoptotic cells in V-RCCS compared to H-RCCS and static controls. Kinetic analysis indicated that apoptosis becomes evident within an hour of seeding of the V-RCCS suggesting an early event in DC generation is affected. However, not all progenitors were susceptible to apoptosis. Furthermore, preliminary studies showed that DC survival in V-RCCS could be enhanced if a dietary nutrient was added to the culture medium at the time of culture initiation. These observations support our hypothesis that nutritional supplements might be effective countermeasures in buffering against the effects of microgravity on DCs.
While the studies described above focus on early events in DC maturation, other investigations have been initiated to explore the mechanisms responsible for the poor functional properties of DC that are generated and survive in the V-RCCS. We have chosen to examine first, the cytoskeletal integrity and migratory capacity of DCs. Cytoskeletal integrity is crucial for various cellular processes including proliferation, adhesion, migration, phagocytosis, cytokine secretion, immune synapse formation and signaling events, to name a few. Thus, we hypothesize that the poor phagocytic capabilities, reduced cytokine secretion and reduced antigen presenting capabilities that we observed in DCs generated in the RCCS could result from a disruption in cytoskeletal components that has been reported previously for cells flown in space or in ground-based models of altered gravity. An alteration in cell structure could also contribute to the reduced generation/survival of DCs in V-RCCS discussed above, and could hinder the ability of DCs to migrate in response to chemokines or other attractants. The latter is an important function of DCs that is currently under intense investigation because of the requirement for DC to migrate to sites of infection/injury/insult for antigen uptake and to sites of lymphocyte accumulation for antigen presentation.

Initial studies suggest that the chemokine receptor CCR7 is upregulated when DCs generated in either control or V-RCCS conditions are matured by exposure to a cocktail of cytokines associated with an inflammatory response. These mature DCs also migrated towards an attractant known to interact with CCR7, i.e. CCL19 chemokine. This would suggest that at least a part of the machinery needed to guide mature DCs to areas where they can interact with lymphocytes (e.g. lymph nodes) remains intact in the altered microgravity conditions. However, we have also noticed that compared to controls, DCs generated and matured within the V-RCCS environment may have a heightened sensitivity to apoptosis and display fewer and shorter dendrites on their surface. These spike-like protrusions of the cell surface membrane appear as DCs mature and, together with increased MHC and costimulatory molecules, enhance the surface area and receptors available for optimal interaction with large numbers of antigen-reactive lymphocytes. These observations could help explain the relative deficiency in antigen presentation that we observed previously for V-RCCS DCs.

NOTE: Greg Lizee has replaced Dr. Savary as PI, effective January 2006 (per info from HQ 9/06). For information after 12/31/2005, search on Lizee and new project title, "Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity."

Bibliography Type:	Description: (Last Updated: 02/27/2014)
Abstracts for Journals and Proceedings	Kulkarni A, Yamauchi K, Sundaresan A, Ambrose C, Wise K, Ramesh G, Pellis N. "Update on the development of a nutritional countermeasure addressing immunology CPR and CRL." NASA Cell Science Conference, January 2005. Abstracts, NASA Cell Science Conference, 2005. , Jan-2005
Abstracts for Journals and Proceedings	Chopra V, Savary CA. "Differentiation of monocytes into endothelial-like precursor cells in microgravity environment." NASA Cell Science Conference, January 2005. Abstracts, NASA Cell Science Conference, 2005. , Jan-2005